CORE VALUES | QUALITY PRODUCTS | SUPERIOR SERVICE

CORE® LIFTING PRODUCTS

WIRE ROPE CLIPS

Crosby®

GP Green Pin®

CM®

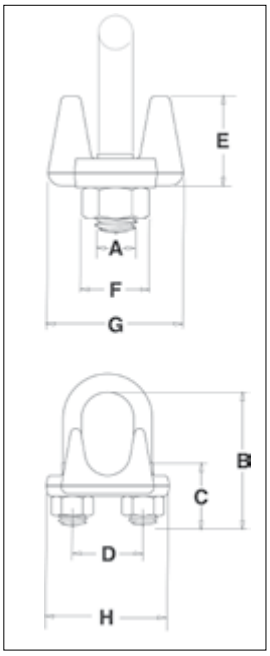
IMPORTED

CoreLifting.com

G-450
Red U-Bolt® Clip

- Each base has a Product Identification Code (PIC) for material traceability, the name Crosby or "CG," and a size forged into it.
- Based on the catalog breaking strength of wire rope, Crosby wire rope clips have an efficiency rating of 80% for 1/8” through 7/8” sizes, and 90% for sizes 1” through 3-1/2”.
- Entire clip is galvanized to resist corrosive and rusting action.
- Sizes 1/8” through 2-1/2” and 3” have forged bases.
- All clips are individually bagged or tagged with proper application instructions and warning information.
- Clip sizes up through 1-1/2” have rolled threads.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these wire rope clips meet other critical performance requirements, including fatigue life, impact properties, and material traceability not addressed by ASME B30.26.
- Look for the Red U-Bolt®, your assurance of genuine Crosby Clips.

APPLICATION AND WARNING INFORMATION
SECTION 17



G-450 Crosby Clips

Rope Size (in)	Rope Size (mm)	Stock No.	Std. Package Qty.	Weight Per 100 (lb)	Dimensions (in) A	B	C	D	E	F	G	H
1/8	3-4*	1010015	100	6	.22	.72	.44	.47	.37	.38	.81	.99
3/16*	5*	1010033	100	10	.25	.97	.56	.59	.50	.44	.94	1.18
1/4	6-7	1010051	100	19	.31	1.03	.50	.75	.66	.56	1.19	1.43
5/16	8	1010079	100	28	.38	1.38	.75	.88	.73	.69	1.31	1.66
3/8	9-10	1010097	100	48	.44	1.50	.75	1.00	.91	.75	1.63	1.94
7/16 - 1/2	11-13	1010131	50	80	.50	1.88	1.00	1.19	1.13	.88	1.91	2.28
9/16 - 5/8	14-16	1010177	50	110	.56	2.25	1.25	1.31	1.34	.94	2.06	2.50
3/4	18-20	1010195	25	142	.62	2.75	1.44	1.50	1.39	1.06	2.25	2.84
7/8	22	1010211	25	212	.75	3.12	1.62	1.75	1.58	1.25	2.44	3.16
1	24-26	1010239	10	252	.75	3.50	1.81	1.88	1.77	1.25	2.63	3.47
1-1/8	28-30	1010257	10	283	.75	3.88	2.00	2.00	1.91	1.25	2.81	3.59
1-1/4	32-34	1010275	10	438	.88	4.44	2.22	2.34	2.17	1.44	3.13	4.13
1-3/8	36	1010293	10	442	.88	4.44	2.22	2.34	2.31	1.44	3.13	4.19
1-1/2	38	1010319	10	544	.88	4.94	2.38	2.59	2.44	1.44	3.41	4.44
1-5/8	41-42	1010337	Bulk	704	1.00	5.31	2.62	2.75	2.66	1.63	3.63	4.75
1-3/4	44-46	1010355	Bulk	934	1.13	5.75	2.75	3.06	2.92	1.81	3.81	5.24
2	48-52	1010373	Bulk	1300	1.25	6.44	3.00	3.38	3.03	2.00	4.44	5.88
2-1/4	56-58	1010391	Bulk	1600	1.25	7.13	3.19	3.88	3.19	2.00	4.56	6.38
2-1/2	62-65	1010417	Bulk	1900	1.25	7.69	3.44	4.13	3.69	2.00	4.69	6.63
** 2-3/4	** 68-72	1010435	Bulk	2300	1.25	8.31	3.56	4.38	4.88	2.00	5.00	6.88
3	75-78	1010453	Bulk	3100	1.50	9.19	3.88	4.75	4.44	2.38	5.31	7.61
** 3-1/2	** 85-90	1010426	Bulk	4000	1.50	10.75	4.50	5.50	6.00	2.38	6.19	8.38

*Electro-plated U-Bolt and Nuts. ** 2-3/4” and 3-1/2” base is made of cast steel.

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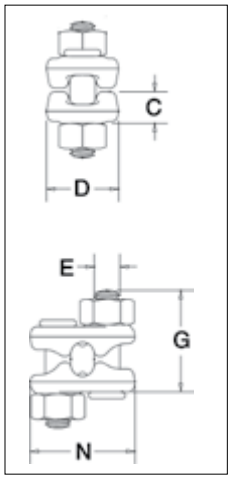
G-429
Fist Grip®Clip
3/16" - 5/8"

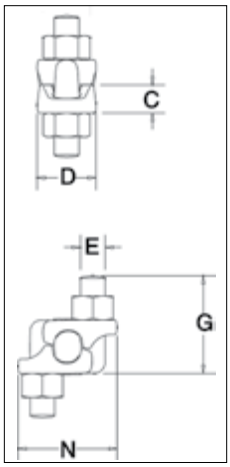
- Entire clip is galvanized to resist corrosive and rusting action.
- Based on the catalog breaking strength of wire rope, Crosby wire rope clips have an efficiency rating of 80% for 3/16" through 7/8" sizes, and 90% for sizes 1" through 1-1/2".
- Bolts are an integral part of the saddle. Nuts can be installed in such a way as to enable the operator to swing the wrench in a full arc for fast installation.
- All sizes have forged steel saddles.
- All Clips are individually bagged or tagged with proper application instructions and warning information.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these wire rope clips meet other critical performance requirements, including fatigue life, impact properties, and material traceability not addressed by ASME B30.26.
- Assembled with standard heavy hex nuts.

G-429
Fist Grip®,Clip
3/4" - 1-1/2"

3/16" - 5/8"



3/4" - 1-1/2"



G-429 Fist Grip® Clips

Rope Size (in)*	Rope Size (mm)	Stock No.	Std. Package Qty.	Weight Per 100 (lb)	Dimensions (in) C	D	E	G	N
3/16 - 1/4	5-7	1010471	100	23	.40	.94	.38	1.41	1.44
5/16	8	1010499	100	28	.47	1.06	.38	1.50	1.54
3/8	10	1010514	50	40	.51	1.06	.44	1.84	1.78
7/16 - 1/2	11-13	1010532	50	62	.59	1.25	.50	2.21	2.15
9/16 - 5/8	14-16	1010550	50	103	.72	1.50	.63	2.72	2.57
3/4	18-20	1010578	25	175	.86	1.81	.75	2.94	2.67
7/8	22	1010596	25	225	.97	2.12	.75	3.31	2.86
1	24-26	1010612	10	300	1.13	2.25	.75	3.72	3.06
1-1/8	28-30	1010630	10	400	1.28	2.38	.88	4.22	3.44
1-1/4	32-34	1010658	10	400	1.34	2.50	.88	4.25	3.56
1-3/8 - 1-1/2	36-40	1010676	Bulk	700	1.56	3.00	1.00	5.56	4.12

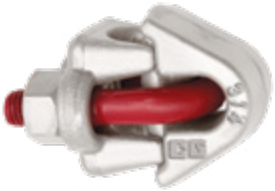
* Sizes through 5/8" incorporate new style design.

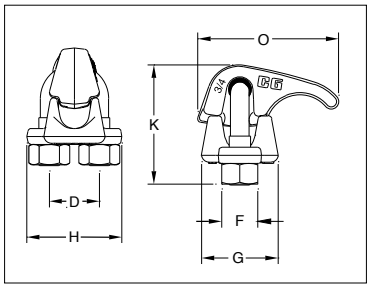
APPLICATION AND WARNING INFORMATION
SECTION 17

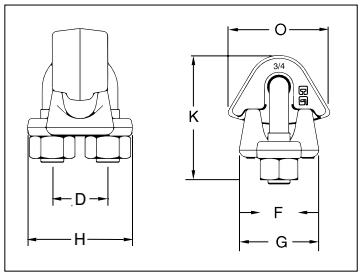
G-460
Soft Eye
Bundle Clip
(For use without Thimble)

- Forged bases and bundle clip adapters.
- All bundle clips are individually bagged or tagged with proper application instructions and warning information.
- Bundle Clip Adapter for Soft Eye (G4460) and for Thimble Eye (G4461) kits available.
- Meets or exceeds all requirements of ASME B30.26 including manufacturing ID and size requirements. Importantly, these wire rope bundle clips meet material traceability not addressed by ASME B30.26.

G-461
Thimble
Eye Bundle Clip







G-460 Soft Eye / G-461 Thimble Eye Bundle Clip

Rope Size (in)	Rope Size (mm)	Bundle Clip Style	Stock No.	Dimensions (in) D	F	G	H	K	O	Weight each (lb)
3/4	18-20	G460	1010509	1.50	1.06	2.25	2.84	3.50	4.13	2.5
3/4	18-20	G461	1010619	1.50	1.06	2.25	2.84	3.50	2.85	2.5

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CROSBY® FIST GRIP® CLIPS

WARNINGS & APPLICATION INSTRUCTIONS

New Style Fist Grip®
3/16" - 5/8"

Fist Grip® Clips
3/4" - 1-1/2"

⚠ WARNING

- **Failure to read, understand, and follow these instructions may cause death or serious injury.**
- **Read and understand these instructions before using clips.**
- **Match the same size clip to the same size wire rope.**
- **Do not mismatch Crosby clips with other manufacturer's clips.**
- **Prepare wire rope end termination only as instructed.**
- **Do not use with plastic coated wire rope.**
- **Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and retighten nuts to recommended torque (See Table 1).**
- **The reuse of clips is discouraged. As recommended by Crosby, have qualified personnel inspect product before use.**

Efficiency ratings for wire rope end terminations are based upon the minimum breaking force of wire rope. The efficiency rating of a properly prepared loop or thimble-eye termination for clip sizes 1/8" through 7/8" is 80%, and for sizes 1" through 3-1/2" is 90%.

The number of clips shown (see Table 1) is based upon using RRL or RLL wire rope, 6 x 19 or 6 x 37 Class, FC or IWRC; IPS or XIP, XXIP. If Seale construction or similar large outer wire type construction in the 6 x 19 Class is to be used for sizes 1 inch and larger, add one additional clip. If a pulley (sheave) is used for turning back the wire rope, add one additional clip.

The number of clips shown also applies to rotation-resistant RRL wire rope, 8 x 19 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller; and to rotation-resistant RRL wire rope, 19 x 7 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller.

For other wire rope manufacture designs not mentioned above, we recommend contacting Crosby Engineering at the address or telephone number on the back cover to ensure the desired efficiency rating.

The style of wire rope termination used for any application is the obligation of the user.

For OSHA (Construction) applications, see OSHA 1926.251.

1. Refer to Table 1 in following these instructions. Turn back specified amount of rope from thimble or loop.

Figure 1

Apply first clip one base width from dead end of rope. Use torque wrench to tighten nuts evenly, alternating from one nut to the other until reaching the recommended torque (See Figure1).

2. When two clips are required, apply the second clip as near the loop or thimble as possible. Use torque wrench to tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. (See Figure 2)

Figure 2

3. When three or more clips are required, space additional clips equally between first two – take up rope slack – use torque wrench to tighten nuts on each clip evenly, alternating from one nut to the other until reaching recommended torque (See Figure 3).

Figure 3

4. If a pulley (sheave) is used in place of a thimble, add one additional Fist Grip. Fist Grip spacing should be as shown. (See Figure 4)

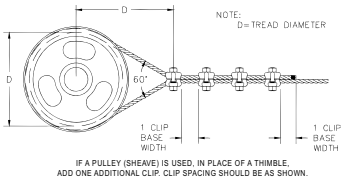

Figure 4

5. WIRE ROPE SPLICING PROCEDURES:

The preferred method of splicing two wire ropes together is to use inter-locking turnback eyes with thimbles, using the recommended number of clips on each eye (See Figure 5).

Figure 5

An alternate method is to use twice the number of clips as used for a turnback termination. The rope ends are placed parallel to each other, overlapping by twice the turnback amount shown in the application instructions. The minimum number of clips should be installed on each dead end (See Figure 6). Spacing, installation torque, and other instructions still apply.

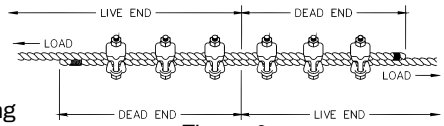

Figure 6

6. IMPORTANT

Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and use torque wrench to retighten nuts to recommended torque.

In accordance with good rigging and maintenance practices, the wire rope end termination should be inspected periodically for wear, abuse, and general adequacy.

Table 1

Clip Size/ Rope Size (in)	(mm)	Minimum No. of Clips	Amount of Rope to Turn Back in Inches	* Torque in ft•lbf
3/16	5	2	4	30
1/4	6-7	2	4	30
5/16	8	2	5	30
3/8	9-10	2	5-1/4	45
7/16	11-12	2	6-1/2	65
1/2	13	3	11	65
9/16	14-15	3	12-3/4	130
5/8	16	3	13-1/2	130
3/4	18-20	3	16	225
7/8	22	4	26	225
1	24-25	5	37	225
1-1/8	28-30	5	41	360
1-1/4	32-34	6	55	360
1-3/8	36	6	62	500
1-1/2	38-40	7	78	500

If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.

If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.

*The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.

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CROSBY® FORGED WIRE ROPE CLIP
WARNINGS & APPLICATION INSTRUCTIONS

G-450
(Red-U-Bolt®)

⚠ WARNING

- **Failure to read, understand, and follow these instructions may cause death or serious injury.**
- **Read and understand these instructions before using clips.**
- **Match the same size clip to the same size wire rope.**
- **Prepare wire rope end termination only as instructed.**
- **Do not use with plastic coated wire rope.**
- **Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and retighten nuts to recommended torque (See Table 1).**
- **The reuse of clips is discouraged. As recommended by Crosby, have qualified personnel inspect product before use.**

Efficiency ratings for wire rope end terminations are based upon the minimum breaking force of wire rope. The efficiency rating of a properly prepared loop or thimble-eye termination for clip sizes 1/8" through 7/8" is 80%, and for sizes 1" through 3-1/2" is 90%.

The number of clips shown (see Table 1) is based upon using RRL or RLL wire rope, 6 x 19 or 6 x 37 Class, FC or IWRC; IPS or XIP, XXIP. If Seale construction or similar large outer wire type construction in the 6 x 19 Class is to be used for sizes 1 inch and larger, add one additional clip. If a pulley (sheave) is used for turning back the wire rope, add one additional clip.

The number of clips shown also applies to rotation-resistant RRL wire rope, 8 x 19 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller; and to rotation-resistant RRL wire rope, 19 x 7 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller.

For other wire rope manufacture designs not mentioned above, we recommend contacting Crosby Engineering at the address or telephone number on the back cover to ensure the desired efficiency rating.

The style of wire rope termination used for any application is the obligation of the user.

For OSHA (Construction) applications, see OSHA 1926.251.

1. Refer to Table 1 following these instructions. Turn back specified amount of rope from thimble or loop. Apply first clip one base width from dead end of rope. Apply U-Bolt over dead end of wire rope – live end rests in saddle (Never saddle a dead horse!). Use torque wrench to tighten nuts evenly, alternate from one nut to the other until reaching the recommended torque (See Figure 1).

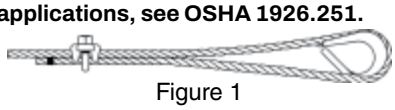

Figure 1

2. When two clips are required, apply the second clip as near the loop or thimble as possible. Use torque wrench to tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. (See Figure 2)

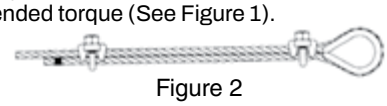

Figure 2

3. When three or more clips are required, space additional clips equally between first two – take up rope slack – use torque wrench to tighten nuts on each clip evenly, alternating from one nut to the other until reaching recommended torque (See Figure 3).

Figure 3

4. If a pulley (sheave) is used in place of a thimble, add one additional clip. Clip spacing should be as shown.

(See Figure 4)

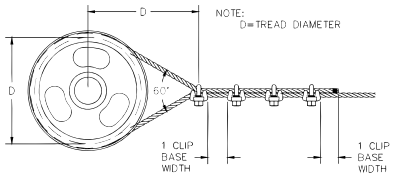

Figure 4

5. WIRE ROPE SPLICING PROCEDURES:

The preferred method of splicing two wire ropes together is to use inter-locking turnback eyes with thimbles, using the recommended number of clips on each eye

(See Figure 5).

Figure 5

An alternate method is to use twice the number of clips as used for a turnback termination. The rope ends are placed parallel to each other, overlapping by twice the turnback amount shown in the application instructions. The minimum number of clips should be installed on each dead end (See Figure 6). Spacing, installation torque, and other instructions still apply.

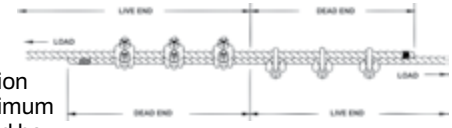

Figure 6

6. IMPORTANT

Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and use torque wrench to retighten nuts to recommended torque.

In accordance with good rigging and maintenance practices, the wire rope end termination should be inspected periodically for wear, abuse, and general adequacy.

Table 1

Rope Size (in)	Rope Size (mm)	Minimum No. of Clips	Amount of Rope to Turn Back in inches	*Torque in ft•lbf
1/8	3-4	2	3-1/4	4.5
3/16	5	2	3-3/4	7.5
1/4	6-7	2	4-3/4	15
5/16	8	2	5-1/4	30
3/8	9-10	2	6-1/2	45
7/16	11-12	2	7	65
1/2	13	3	11-1/2	65
9/16	14-15	3	12	95
5/8	16	3	12	95
3/4	18-20	4	18	130
7/8	22	4	19	225
1	24-25	5	26	225
1-1/8	28-30	6	34	225
1-1/4	33-34	7	44	360
1-3/8	36	7	44	360
1-1/2	38-40	8	54	360
1-5/8	41-42	8	58	430
1-3/4	44-46	8	61	590
2	48-52	8	71	750
2-1/4	56-58	8	73	750
2-1/2	62-65	9	84	750
2-3/4	68-72	10	100	750
3	75-78	10	106	1200
3-1/2	85-90	12	149	1200

If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.

If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.

*The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.

This application/warning information apply to Crosby products only.

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LIFTING SLING FITTINGS FOR WIRE ROPE

WIRE ROPE CLIPS

Applications

Wire rope clips are used on wire rope eye-loop connections and on complete loops. They are used in end-to-end connections where socketing or splicing is not feasible and when a temporary joint is required.

More information about the range, certification and finish, please go to the introduction of chapter 3 lifting slings.

Design

Green Pin® wire rope clips are drop forged and have a bridge with grooves to tighten the wire rope properly in the clip; the DIN wire rope clips have a malleable base, without grooves.

Wire rope clips are generally marked with:

- manufacturer's symbol - e.g. GP
- wire rope diameter in mm or inches - e.g. 13 or $^1/_2$"
- traceability code - e.g. A1

Instructions for use

Wire rope clips should be inspected before use to ensure that:

- all markings are legible;
- a wire rope clip with the correct dimensions has been selected;
- the nuts or any other locking system cannot vibrate out of position;
- the wire rope clip is free from nicks, gouges and cracks;

Never modify, repair or reshape a wire rope clip by machining, welding, heating or bending as this may affect their performance.

The wire rope clip should be fitted to the wire rope as shown in below figures. The bridge of the wire rope clip should always be placed on the load bearing part of the rope. The U-bolt of the clip should be placed on the rope tail, also known as the dead end of the rope. Turn back enough wire rope length so that the required minimum number of clips can be installed according to the instructions below.

The first clip must be placed one bridge width from the turned-back rope tail or dead end of the rope, according to figure 1. Tighten the nuts to the specified torque.

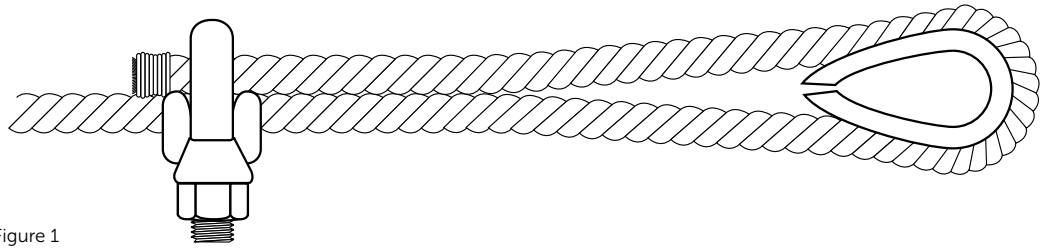


Figure 1

The second clip must be placed immediately against the thimble. Take care that the correct tightening of the clip does not damage the outer wires of the wire rope (figure 2). Tighten the nuts firmly but not yet to the specified torque.

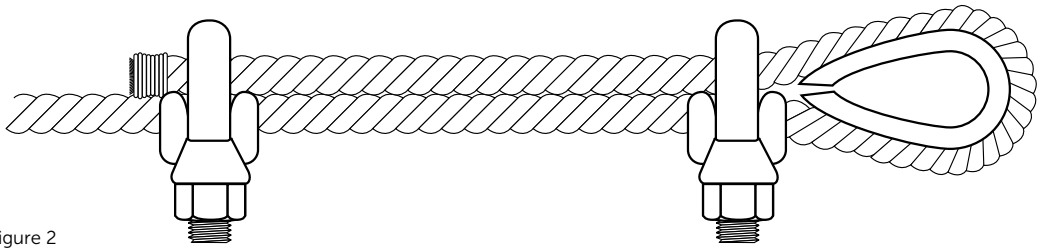


Figure 2

The following clips should be placed on the wire rope between the first and second clip in such a way that they are separated by at least 1½ times the clip-width with a maximum of 3 times the clip-width, according to figure 3.

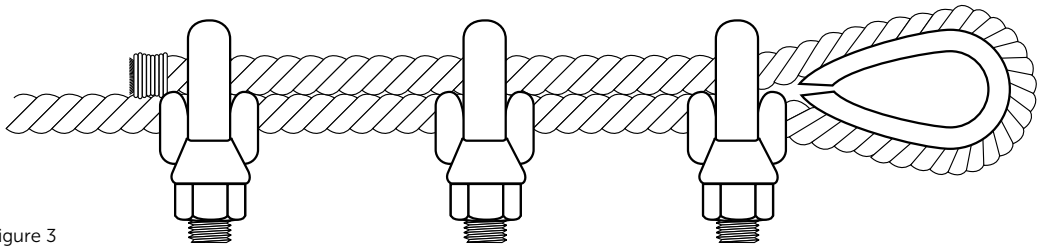


Figure 3

Apply light tension on the rope and tighten all nuts evenly, alternating until reaching the specified torque. After assembly and before the rope is taken into service, the nuts must be tightened further to the prescribed torque. After the load has been applied to the assembly for the first time, the torque value must be checked and corrected if necessary. Re-tightening of the nuts must be done at 10.000 cycles (heavy usage), 20.000 cycles (moderate usage) or 50.000 cycles (light usage). If cycles are unknown, a competent person could fix a time period, e.g. every 3 months, 6 months, annually.

The torque values and the minimum number of clips to be applied for a particular rope size are given in the following tables.

diameter wire rope	diameter wire rope	min. no of clips required	length of rope to turn back	torque value	torque value
inch	inch		inch	Ft.Lbs	Ft.Lbs
1/8	3 - 4	2	85	6.1	4.5
3/16	5	2	95	10.2	7.5
1/4	6 - 7	2	120	20.3	15
5/16	8	3	133	40.7	30
3/8	9 - 10	3	165	61	45
7/16	11	3	178	88	65
1/2	12 - 13	3	292	88	65
9/16	14 - 15	3	305	129	95
5/8	16	3	305	129	95
3/4	18 - 20	4	460	176	130
7/8	22	4	480	305	225
1	24 - 26	5	660	305	225
1 1/8	28 - 30	6	860	305	225
1 1/4	32 - 34	7	1120	488	360
1 3/8	36	7	1120	488	360
1 1/2	38 - 40	8	1370	488	360
1 5/8	41 - 42	8	1470	583	430
1 3/4	44 - 46	8	1550	800	590
2	48 - 52	8	1800	1017	750
2 1/4	56 - 58	8	1850	1017	750
2 1/2	62 - 65	9	2130	1017	750
2 3/4	68 - 72	10	2540	1017	750
3	75 - 78	10	2690	1627	1200

Table 1, Green Pin® wire rope clips generally to EN 13411-5 Type B, required number and torque value

diameter wire rope	min. no of clips required	torque value	torque value
inch		Nm	Ft.Lbs
5	3	2	1.5
6.5	3	3.5	2.6
8	4	6	4.4
10	4	9	6.6
12	4	20	14.8
13	4	33	24.3
14	4	33	24.3
16	4	49	36
19	5	68	50
22	5	107	79
26	5	147	108
30	6	212	156
34	6	296	218
40	6	363	268

Table 2, Wire rope clips generally to EN 13411-5 Type A, required number and torque value

The efficiency of a wire rope termination made with wire rope clips depends on the correct placement of the clips on the rope and on correct fitting and tightening of the clips. With inadequately tightened nuts or with an insufficient number of wire rope clips the rope end may slide through the clips during use.

The fitting of the clips on the ropes may be affected by various circumstances, such as:
- the nut may be tight on the thread, yet not tight against the bridge;
- contamination of the thread by dirt, oil or corrosion products, which may prevent correct tightening of the nut.

Forged wire rope clips provide greater bearing surface and more consistent strength than malleable cast iron clips.

Suitable applications of wire rope clips to EN 13411-5 standards include suspending static loads and single use lifting operations which have been assessed by a competent person taking into account appropriate safety factors.

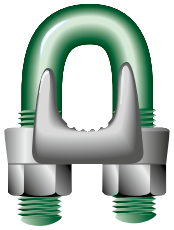
Wire rope clips should not be used in following applications:
- hoist ropes in mines;
- rope drives for cranes in steel works and rolling mills;
- permanent fastening of ropes in other rope drives;
- rope terminations for load suspension devices in the operation of lifting appliances, except in the case of lifting tackles where these are produced for a special application and used only once.

Wire rope clips must be regularly inspected in accordance with the safety standards given in the country of use. This is required because the products in use may be affected by wear, misuse, overloading etc. which may lead to deformation and alteration of the material structure. Inspection should take place at least every six months (follow the local rules in the country of use) and more frequently when the products are used in severe operating conditions.

Green Pin® Wire Rope Clip

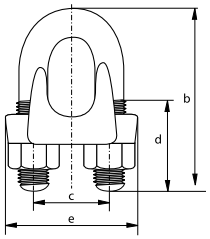
Wire rope clip generally to EN 13411-5 Type B

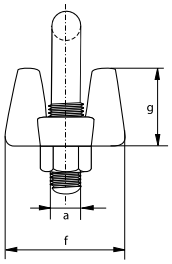
Scan for additional product details



- **Material:** bridge: drop forged high tensile steel SAE 1045
  U-bolt: SAE 1015
- **Standard:** EN 13411-5 Type B
  formerly U.S. Federal Specification FF-C-450D
- **Finish:** hot dipped galvanized
  U-bolt and/or nuts for diameter bow $^{3}/_{16}$ inch, $^{1}/_{4}$ inch, $^{5}/_{16}$ inch and $^{13}/_{32}$ inch are electro-galvanized
- **Certification:** 2.1 2.2 CE IIB
- **Article code:** scan QR code to see article codes

G-6240





diameter wire rope	diameter	length bow	width inside	length thread	length base	thickness base	height base	weight per 100 pcs
	a	b	c	d	e	f	g	
inch	inch	inch	inch	inch	inch	inch	inch	lbs
$^{1}/_{8}$	$^{3}/_{16}$	$^{15}/_{16}$	$^{15}/_{32}$	$^{7}/_{16}$	$^{15}/_{16}$	$^{13}/_{16}$	$^{13}/_{32}$	6.61
$^{3}/_{16}$	$^{1}/_{4}$	$1\,^{7}/_{32}$	$^{19}/_{32}$	$^{1}/_{2}$	$1\,^{5}/_{32}$	$^{15}/_{16}$	$^{1}/_{2}$	15.43
$^{1}/_{4}$	$^{5}/_{16}$	$1\,^{11}/_{32}$	$^{3}/_{4}$	$^{1}/_{2}$	$1\,^{15}/_{32}$	$1\,^{3}/_{16}$	$^{23}/_{32}$	18.52
$^{5}/_{16}$	$^{13}/_{32}$	$1\,^{25}/_{32}$	$^{7}/_{8}$	$^{3}/_{4}$	$1\,^{11}/_{16}$	$1\,^{5}/_{16}$	$^{3}/_{4}$	27.3
$^{3}/_{8}$	$^{7}/_{16}$	$1\,^{15}/_{16}$	$1\,^{1}/_{32}$	$^{3}/_{4}$	$1\,^{15}/_{16}$	$1\,^{21}/_{32}$	$^{31}/_{32}$	46.3
$^{7}/_{16}$	$^{15}/_{32}$	$2\,^{3}/_{8}$	$1\,^{3}/_{16}$	$^{31}/_{32}$	$2\,^{9}/_{32}$	$1\,^{13}/_{16}$	$1\,^{1}/_{32}$	73
$^{1}/_{2}$	$^{1}/_{2}$	$2\,^{13}/_{32}$	$1\,^{3}/_{16}$	$^{31}/_{32}$	$2\,^{9}/_{32}$	$1\,^{7}/_{8}$	$1\,^{7}/_{32}$	73
$^{9}/_{16}$	$^{9}/_{16}$	$2\,^{27}/_{32}$	$1\,^{5}/_{16}$	$1\,^{1}/_{4}$	$2\,^{15}/_{32}$	$2\,^{1}/_{16}$	$1\,^{7}/_{32}$	101
$^{5}/_{8}$	$^{9}/_{16}$	$2\,^{29}/_{32}$	$1\,^{5}/_{16}$	$1\,^{1}/_{4}$	$2\,^{17}/_{32}$	$2\,^{1}/_{8}$	$1\,^{13}/_{32}$	101
$^{3}/_{4}$	$^{5}/_{8}$	$3\,^{3}/_{8}$	$1\,^{1}/_{2}$	$1\,^{15}/_{32}$	$2\,^{27}/_{32}$	$2\,^{1}/_{4}$	$1\,^{1}/_{2}$	142
$^{7}/_{8}$	$^{3}/_{4}$	$3\,^{27}/_{32}$	$1\,^{25}/_{32}$	$1\,^{5}/_{8}$	$3\,^{5}/_{32}$	$2\,^{7}/_{16}$	$1\,^{9}/_{16}$	213
1	$^{3}/_{4}$	$4\,^{1}/_{4}$	$1\,^{7}/_{8}$	$1\,^{13}/_{16}$	$3\,^{15}/_{32}$	$2\,^{5}/_{8}$	$1\,^{27}/_{32}$	254
$1\,^{1}/_{8}$	$^{3}/_{4}$	$4\,^{19}/_{32}$	2	2	$3\,^{19}/_{32}$	$2\,^{7}/_{8}$	$1\,^{7}/_{8}$	280
$1\,^{1}/_{4}$	$^{7}/_{8}$	$5\,^{1}/_{8}$	$2\,^{5}/_{16}$	$2\,^{1}/_{8}$	$4\,^{1}/_{8}$	$3\,^{1}/_{8}$	$2\,^{7}/_{32}$	434
$1\,^{3}/_{8}$	$^{7}/_{8}$	$5\,^{1}/_{2}$	$2\,^{3}/_{8}$	$2\,^{5}/_{16}$	$4\,^{1}/_{4}$	$3\,^{1}/_{8}$	$2\,^{9}/_{32}$	454
$1\,^{1}/_{2}$	$^{7}/_{8}$	$5\,^{25}/_{32}$	$2\,^{19}/_{32}$	$2\,^{3}/_{8}$	$4\,^{13}/_{32}$	$3\,^{11}/_{32}$	$2\,^{17}/_{32}$	560
$1\,^{5}/_{8}$	$^{31}/_{32}$	$6\,^{11}/_{32}$	$2\,^{3}/_{4}$	$2\,^{5}/_{8}$	$4\,^{3}/_{4}$	$3\,^{5}/_{8}$	$2\,^{5}/_{8}$	710
$1\,^{3}/_{4}$	$1\,^{5}/_{32}$	$6\,^{27}/_{32}$	$3\,^{1}/_{16}$	$2\,^{3}/_{4}$	$5\,^{9}/_{32}$	$3\,^{13}/_{16}$	3	922
2	$1\,^{1}/_{4}$	$7\,^{11}/_{16}$	$3\,^{3}/_{8}$	$3\,^{1}/_{16}$	$5\,^{29}/_{32}$	$4\,^{7}/_{16}$	$3\,^{11}/_{32}$	1327
$2\,^{1}/_{4}$	$1\,^{1}/_{4}$	$8\,^{3}/_{8}$	$3\,^{27}/_{32}$	$3\,^{3}/_{16}$	$6\,^{3}/_{8}$	$4\,^{9}/_{16}$	$3\,^{15}/_{16}$	1711
$2\,^{1}/_{2}$	$1\,^{1}/_{4}$	$8\,^{15}/_{16}$	$4\,^{1}/_{8}$	$3\,^{7}/_{16}$	$6\,^{5}/_{8}$	$4\,^{11}/_{16}$	$4\,^{7}/_{16}$	1900
$2\,^{3}/_{4}$	$1\,^{1}/_{4}$	$9\,^{9}/_{16}$	$4\,^{13}/_{32}$	$3\,^{19}/_{32}$	$6\,^{27}/_{32}$	5	$4\,^{7}/_{8}$	2238
3	$1\,^{1}/_{2}$	$10\,^{21}/_{32}$	$4\,^{3}/_{4}$	$3\,^{27}/_{32}$	$7\,^{5}/_{8}$	$5\,^{5}/_{16}$	$5\,^{11}/_{32}$	2804

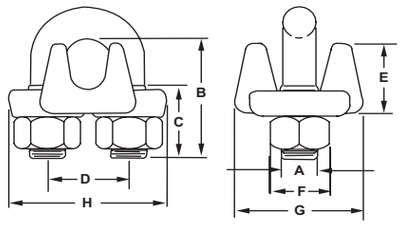
WIRE ROPE CLIPS

MADE IN THE USA

SIZES: 1/8 TO 1-1/2 IN.

BENEFITS & FEATURES

- Meets the dimensional and finish requirements of FF-C-450
- Drop forged saddle
- Forged saddles include size, USA and CM logo
- Drop-forged base for superior function and toughness
- Product Codes with a -2 suffix are packed and priced two per package.

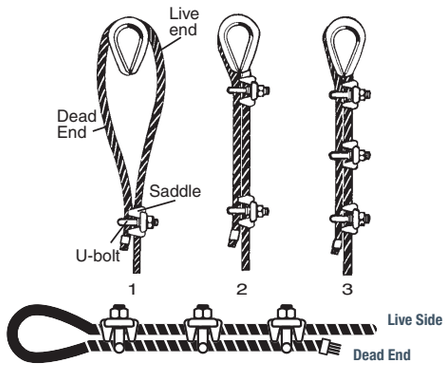


Size (in.)	Standard Package	Product Code	Dimensions (in.)								Minimum Number of Clips	Torque (ft./lbs.)	Rope Turnback	Weight (lbs.)
			A	B	C	D	E	F	G	H				
1/8	50	M244-2	1/8-24UNC	0.72	0.44	0.47	0.42	0.38	0.81	1.09	2	4-1/2	3-1/4	0.08
3/16	50	M245-2	1/4-20UNC	0.94	0.56	0.59	0.50	0.50	0.94	1.19	2	7-1/2	3-3/4	0.12
1/4	50	M246-2	5/16-18UNC	1.03	0.50	0.75	0.66	0.56	1.19	1.44	2	15	4-3/4	0.18
5/16	50	M247-2	3/8-16UNC	1.38	0.75	0.88	0.72	0.69	1.31	1.69	2	30	5-1/4	0.30
3/8	50	M248-2	7/16-14UNC	1.50	0.75	1.00	0.91	0.75	1.63	1.94	2	45	6-1/2	0.42
7/16	50	M249-2	1/2-13UNC	1.88	1.00	1.19	1.06	0.88	1.78	2.31	2	65	7	0.70
1/2	20	M250	1/2-13UNC	1.88	1.00	1.19	1.16	0.88	1.91	2.31	3	65	11-1/2	0.85
9/16	20	M296	9/16-12UNC	2.38	1.25	1.31	1.24	0.94	1.94	2.50	3	95	12	1.00
5/8	20	M251	9/16-12UNC	2.38	1.25	1.31	1.34	0.94	2.00	2.49	3	95	12	1.00
3/4	10	M252	5/8-11UNC	2.75	1.44	1.50	1.44	1.06	2.34	2.81	4	130	18	1.53
7/8	10	M253	3/4-10UNC	3.13	1.63	1.75	1.63	1.25	2.44	3.16	4	225	20	2.40
1	10	M254	3/4-10UNC	3.50	1.81	1.88	1.78	1.25	2.63	3.47	5	225	26	2.50
1-1/8	5	M255	3/4-10UNC	3.88	2.00	2.00	1.88	1.25	2.81	3.59	6	225	34	3.10
1-1/4	5	M256	7/8-9UNC	4.25	2.13	2.31	2.19	1.44	3.18	4.22	7	360	37	4.10
1-3/8	5	M257	7/8-9UNC	4.63	2.31	2.38	2.25	1.44	3.08	4.25	7	360	44	4.50
1-1/2	5	M258	7/8-9UNC	4.94	2.38	2.59	2.50	1.44	3.41	4.47	8	360	48	5.40

NOTE: 1/8" through 5/8" packed 1 piece per poly bag. 3/4" & larger shipped assembled in factory packs and tagged.

PROPER USE OF WIRE ROPE CLIPS

1. Refer to the chart above when following these instructions. Turn back specified amount of rope from thimble or loop. Apply first clip one base width from dead end of rope. Apply U-bolt over dead end of wire rope, ensuring live end rests in saddle. Tighten nuts evenly, alternate from one nut to the other until reaching the recommended torque.
2. When two clips are required, apply the second clip as near the thimble or loop as possible. Tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible and turn nuts on second clip firmly, but do not tighten. Proceed to Step 3.
3. When three or more clips are required, space additional clips equally between the first two and take up rope slack. Then tighten nuts on each U-bolt evenly, alternating from one nut to the other until reaching the recommended torque.
4. Apply an initial load equal to loads expected in use. Inspect for proper orientation and spacing of clips and retighten the nuts to recommended torque.



NOTE: Mechanical spliced or flemished eyes slings are the preferred method of wire rope sling construction. OSHA does not allow the use of clips to form the eyes of wire rope slings.

CARE

- Care should be exercised in the installation and use of wire rope clips so that the clip, wire rope, or thimble is not damaged.
- Do not over torque or under torque the nuts. Too much torque can result in damage to the clip and/or the wire rope. Too little torque can result in the wire rope slipping. Torque nuts to the value specified in the accompanying instructions.
- Clips should not be subjected to bending or come in contact with sharp object.
- Avoid exposure to corrosive mediums.

INSPECTION

- Visually inspect wire rope clips before each use.
- Be certain threads are not stripped and that nuts are tight.
- Check torque of nuts periodically.
- Replace distorted thimbles.
- Shorten wire rope and form new loop if damaged.
- Replace distorted thimbles.

WARNING

NEVER INSTALL THE U-BOLT ON THE LIVE SIDE!

IMPROPER USE OF WIRE ROPE CLIPS

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MID-GRIP WIRE ROPE CLIPS

MADE IN THE USA

SIZES: 3/16 TO 3/4 IN.

BENEFITS & FEATURES

- Allows for full arc wrench swing for quicker installation, retightening and disassembly.
- Saddles are made of forged steel for strength and toughness. Full assembly is mechanical galvanized per ASTM B695-04 Class 25 Type 1.
- Size, manufacturer and trace code are clearly marked on the saddle for easy identification.
- Meets or exceeds performance requirements of FF-C-450 specifications, Type III, Class 1 and will provide maximum holding strength.
- Product Codes with a -2 suffix are packed and priced two per package.

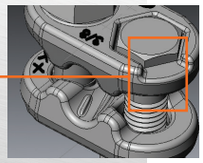
NEW & IMPROVED HEX BOLT & NUT DESIGN

TIGHT, SECURE FIT
Hexagon bolt head fits securely into hex-shaped socket for exceptional rotation resistance. This prevents spinning even after repeated use and re-torqueing.

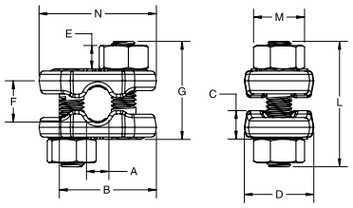
PRECISE THREADING
Precise threading on bolt and nut ensures tight alignment and exceptional strength.

FEWER LOOSE PARTS TO DROP
Hex bolt is knurled & machine pressed into socket to securely couple with the saddle.

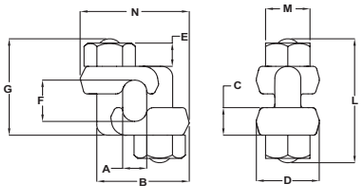
EASY TO INSPECT
Sight window/notch on top of mid-grip allows for easy inspection of the bolts.



NEW STYLE
Available in 3/16" through 5/8"



OLD STYLE
Available in 3/4" only

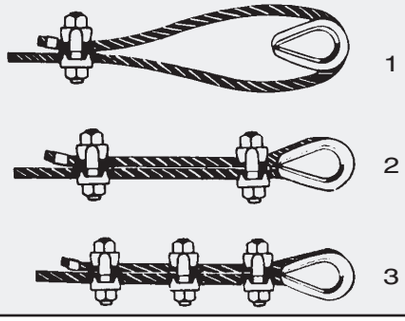


Size (in.)	Standard Package	Product Code	Dimensions (in.)										Minimum Number of Clips	Rope Turn-Back (in.)	Torque (ft./lbs.)	Weight (lbs.)
			A	B	C	D	E Thread	F	G	L Approx.	M	N				
New Style																
3/16	20	M2546-2	0.37	1.53	0.44	1.06	3/8 - 16	0.50	1.50	1.93	0.69	1.74	2	4	30	0.38
1/4	20	M2546-2	0.37	1.53	0.44	1.06	3/8 - 16	0.50	1.50	1.93	0.69	1.74	2	4	30	0.38
5/16	20	M2547-2	0.34	1.48	0.44	1.06	3/8 - 16	0.63	1.50	1.90	0.69	1.80	2	5	30	0.38
3/8	20	M2548-2	0.46	1.67	0.54	1.13	7/16 - 14	0.75	1.84	2.26	0.75	2.00	2	5.5	45	0.49
7/16	20	M2550	0.65	2.03	0.59	1.31	1/2 - 13	1.05	2.23	2.74	0.88	2.40	2	6.5	65	0.84
1/2	20	M2550	0.65	2.03	0.59	1.31	1/2 - 13	1.05	2.23	2.74	0.88	2.40	3	11	65	0.84
9/16	20	M2551	0.78	2.46	0.73	1.63	5/8 - 11	1.38	2.81	3.44	1.06	2.90	3	12.75	130	1.55
5/8	20	M2551	0.78	2.46	0.73	1.63	5/8 - 11	1.38	2.81	3.44	1.06	2.90	3	13.5	130	1.55
Old Style																
3/4	20	M2252	0.81	2.69	0.88	1.81	3/4 - 10	1.50	3.38	5.00	1.25	3.02	3	16	225	1.79

INSPECTION, CARE & USE

PROPER USE OF MID-GRIP CLIPS

1. Refer to chart above when following these instructions. Turn back specified amounts of rope from thimble or loop. Apply first clip one base width from dead end of rope. Tighten nuts evenly, alternating from one nut to the other until reaching the recommended torque.
2. When two clips are required, apply the second clip as near the loop or thimble as possible. Tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible and turn nuts on second clip firmly, but do not tighten. Proceed to step 3.
3. When three or more clips are required, space additional clips between first two and take up rope slack. Then tighten nuts on all clips, alternating from one nut to the other until reaching the recommended torque.
4. Apply an initial load equal to loads expected in use. Inspect for proper spacing and retighten the nuts to the recommended torque.



For more information, visit us at
www.cmco.com

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MID-GRIP WIRE ROPE CLIPS

MADE IN THE USA

SIZES: 3/16 TO 3/4 IN.

BENEFITS & FEATURES

- Allows for full arc wrench swing for quicker installation, retightening and disassembly.
- Saddles are made of forged steel for strength and toughness. Full assembly is mechanical galvanized per ASTM B695-04 Class 25 Type 1.
- Size, manufacturer and trace code are clearly marked on the saddle for easy identification.
- Meets or exceeds performance requirements of FF-C-450 specifications, Type III, Class 1 and will provide maximum holding strength.
- Product Codes with a -2 suffix are packed and priced two per package.

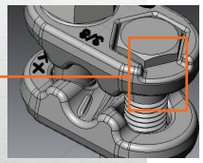
NEW & IMPROVED HEX BOLT & NUT DESIGN

TIGHT, SECURE FIT
Hexagon bolt head fits securely into hex-shaped socket for exceptional rotation resistance. This prevents spinning even after repeated use and re-torqueing.

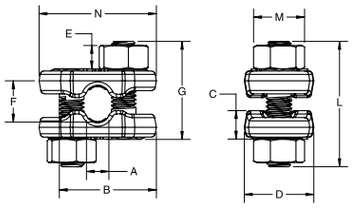
PRECISE THREADING
Precise threading on bolt and nut ensures tight alignment and exceptional strength.

FEWER LOOSE PARTS TO DROP
Hex bolt is knurled & machine pressed into socket to securely couple with the saddle.

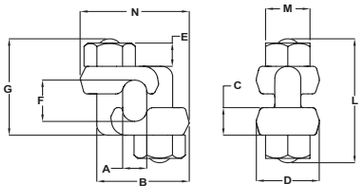
EASY TO INSPECT
Sight window/notch on top of mid-grip allows for easy inspection of the bolts.



NEW STYLE
Available in 3/16" through 5/8"



OLD STYLE
Available in 3/4" only

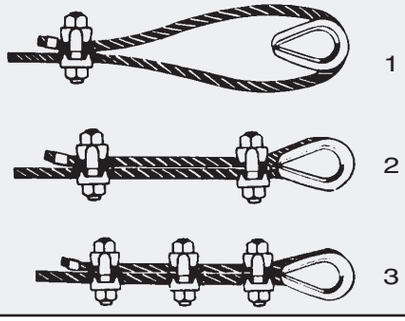


Size (in.)	Standard Package	Product Code	Dimensions (in.)										Minimum Number of Clips	Rope Turn-Back (in.)	Torque (ft./lbs.)	Weight (lbs.)
			A	B	C	D	E Thread	F	G	L Approx.	M	N				
New Style																
3/16	20	M2546-2	0.37	1.53	0.44	1.06	3/8 - 16	0.50	1.50	1.93	0.69	1.74	2	4	30	0.38
1/4	20	M2546-2	0.37	1.53	0.44	1.06	3/8 - 16	0.50	1.50	1.93	0.69	1.74	2	4	30	0.38
5/16	20	M2547-2	0.34	1.48	0.44	1.06	3/8 - 16	0.63	1.50	1.90	0.69	1.80	2	5	30	0.38
3/8	20	M2548-2	0.46	1.67	0.54	1.13	7/16 - 14	0.75	1.84	2.26	0.75	2.00	2	5.5	45	0.49
7/16	20	M2550	0.65	2.03	0.59	1.31	1/2 - 13	1.05	2.23	2.74	0.88	2.40	2	6.5	65	0.84
1/2	20	M2550	0.65	2.03	0.59	1.31	1/2 - 13	1.05	2.23	2.74	0.88	2.40	3	11	65	0.84
9/16	20	M2551	0.78	2.46	0.73	1.63	5/8 - 11	1.38	2.81	3.44	1.06	2.90	3	12.75	130	1.55
5/8	20	M2551	0.78	2.46	0.73	1.63	5/8 - 11	1.38	2.81	3.44	1.06	2.90	3	13.5	130	1.55
Old Style																
3/4	20	M2252	0.81	2.69	0.88	1.81	3/4 - 10	1.50	3.38	5.00	1.25	3.02	3	16	225	1.79

INSPECTION, CARE & USE

PROPER USE OF MID-GRIP CLIPS

1. Refer to chart above when following these instructions. Turn back specified amounts of rope from thimble or loop. Apply first clip one base width from dead end of rope. Tighten nuts evenly, alternating from one nut to the other until reaching the recommended torque.
2. When two clips are required, apply the second clip as near the loop or thimble as possible. Tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible and turn nuts on second clip firmly, but do not tighten. Proceed to step 3.
3. When three or more clips are required, space additional clips between first two and take up rope slack. Then tighten nuts on all clips, alternating from one nut to the other until reaching the recommended torque.
4. Apply an initial load equal to loads expected in use. Inspect for proper spacing and retighten the nuts to the recommended torque.



For more information, visit us at
www.cmco.com

RG4761_Rigging Catalog Final.indd 108 7/2/25 12:02 PM

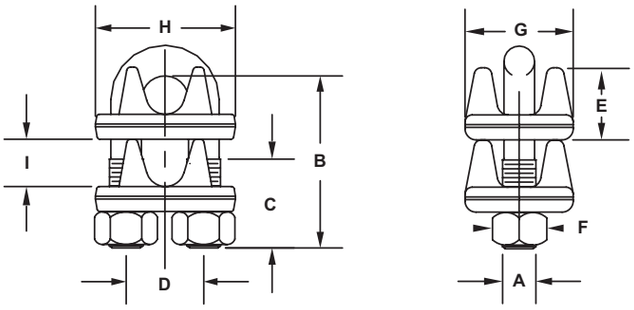
PIGGYBACK® WEDGE SOCKET CLIPS

MADE IN THE USA

SIZES: 3/8 TO 1-1/2 IN.

BENEFITS & FEATURES

- Forged saddles include size, USA and CM logo
- Drop-forged base for superior function and toughness
- Galvanized finish



Size (in.)	Standard Package	Product Code	Dimensions (in.)									Torque (ft./lbs.)	Min. Tail (in.)	Weight (lbs.)
			A	B	C	D	E	F	G	H	I			
3/8	10	M248D	7/16-14UNC	2.38	1.00	1.00	0.53	0.75	1.62	1.94	0.53	45	6	0.8
7/16	10	M249D	1/2-13UNC	2.62	1.00	1.19	0.66	0.88	1.78	2.31	0.66	65	6	1.3
1/2	10	M250D	1/2-13UNC	2.62	1.00	1.19	0.72	0.88	1.91	2.31	0.72	65	6	1.4
9/16	10	M296D	9/16-12UNC	3.19	1.25	1.31	0.77	0.94	1.97	2.50	0.77	95	6	1.7
5/8	10	M251D	9/16-12UNC	3.19	1.25	1.31	0.87	0.94	2.00	2.50	0.87	95	6	1.7
3/4	10	M252D	5/8-11UNC	3.50	1.50	1.50	0.91	1.06	2.34	2.81	0.91	130	6	2.5
7/8	5	M253D	3/4-10UNC	3.88	2.00	1.75	1.07	1.25	2.44	3.16	1.07	225	6	3.6
1	5	M254D	3/4-10UNC	4.25	2.00	1.88	1.22	1.25	2.62	3.47	1.22	225	6	3.9
1-1/8	5	M255D	3/4-10UNC	4.50	2.00	2.00	1.31	1.25	2.81	3.59	1.31	225	6-3/4	4.9
1-1/4	5	M256D	7/8-9UNC	5.25	2.38	2.31	1.50	1.44	3.18	4.22	1.50	360	7-1/2	6.5
1-3/8	5	M257D	7/8-9UNC	5.62	2.38	2.38	1.56	1.44	3.08	4.25	1.56	360	8-1/4	7.1
1-1/2	5	M258D	7/8-9UNC	6.00	2.38	2.59	1.75	1.44	3.41	4.47	1.75	360	9	8.2

PROPER USE OF PIGGYBACK WEDGE SOCKET CLIPS

- Dead end wire rope tail length should be at least 6 rope diameters, but not less than 6 inches beyond the wedge socket. See Figure 1.
- Apply U-bolt and first saddle on dead end rope and the second saddle on live end rope. Install nuts. See Figures 1 & 2.
- Position the dual saddle clip so that 3 to 4 inches of dead end rope remain beyond the clip. See Figures 1 & 2.
- Tighten nuts evenly. Alternate tightening each nut until reaching the required torque specified. See Figure 2.
- Check that clip does not pinch or clamp the live wire rope. Apply an initial test load equal to the loads expected in regular use. Be certain rope is secured properly in the wedge socket prior to any use.
- Inspect for proper rope alignment with wedge socket and retighten the nuts to the specified torque.

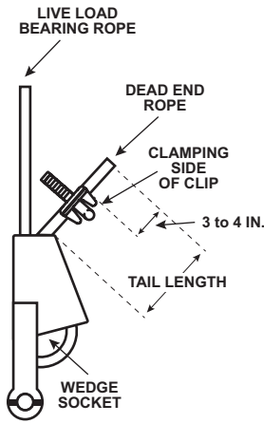


FIGURE 1

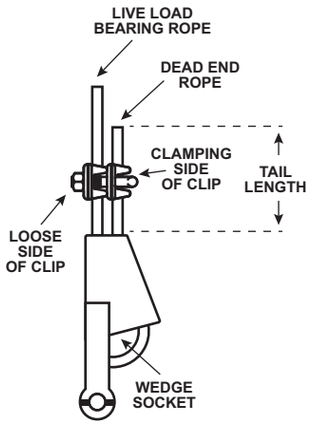


FIGURE 2

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WIRE ROPE CLIPS - WARNINGS AND INFO.

Read and follow ALL instructions and warnings. Failure to do so could result in injury or death.

Select the right wire rope clip for your application

- NEVER use cast malleable wire rope clips for lifting or suspending ANY load.
- Match the same size wire rope clip to the same size rope diameter. For example, use a 3/8" wire rope clip with a 3/8" wire rope.
- When working with coated cable, ALWAYS:
  - match wire rope clip size to uncoated cable diameter and
  - strip coating off cable where wire rope clips will be attached.

Install wire rope clips correctly

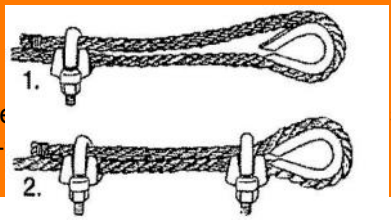
- Never use fewer than the number of wire rope clips defined in the tables on the next three pages.
- Wire rope clips are made for use with wire rope only. Never use wire rope clips on pipe, rod, chain, fiber rope, etc.

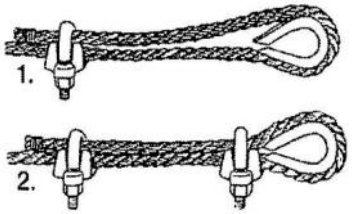
Use wire rope clips properly

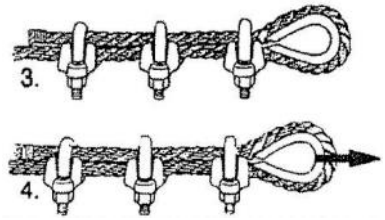
- Never connect two ends of wire rope with wire rope clips.
- Never shock or impact load.

Inspect and maintain wire rope clips regularly

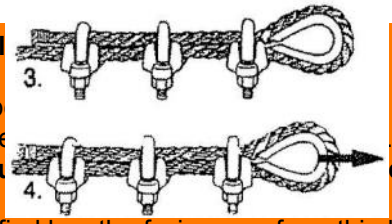
- Inspect wire rope clips before each use. Discard and replace storted, or damaged.
- Inspect wire rope clip nuts before and after each use and re-







Correct Method of I



When using wire rop
exercised to make ce
Failure to do so cou eath.

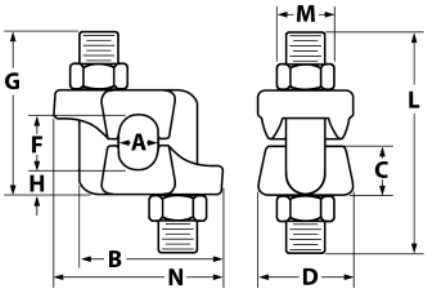
1. Turn back specified length of wire rope from thimble (see tables on next three pages). Put first clip one saddle width from seized "dead end" (Fig. 1). Seat "live end" (load carrying part) of wire rope in saddle and position U-bolt over "dead end." Remember: "**Never saddle a dead horse**." Tighten nuts evenly to proper torque (see tables on next five pages).
2. Put second clip close to the thimble without binding it (Fig. 2). Install nuts firmly but do not yet tighten to proper torque.
3. Install all remaining clips equally spaced between the first two clips (Fig. 3). Install nuts firmly but do not yet tighten to proper torque.
4. Apply light tension to wire rope assembly, and then tighten all nuts evenly to proper torque (Fig. 4).
5. Inspect wire rope clip nuts before and after each use and re-torque as needed. When loads are applied, the wire rope will stretch slightly causing the rope to slip and the termination to come loose, which may cause serious injury or death.

Add at least one more clip if a pulley (sheave) is used in place of a thimble. If uncertain, check with the wire rope manufacturer. If more clips are used than specified, proportionally increase the amount of wire rope that is turned back.

When the required minimum number of clips is installed in accordance with these written instructions, they will develop up to 80% efficiency when used on right lay wire rope of classes 6x19, 6x36, 7x19, 8x19, 19x7, and cable laid. Drop forged wire rope clips are recommended for use only with steel core wire ropes.

WARNING: NEVER EXCEED WORKING LOAD LIMIT

DOUBLE K GRIP WIRE ROPE CLIPS



GALVANIZED, DROP FORGED

Dimensions in Inches						Min. # Clips	Wire Rope Turn Back in Inches	Torque* (ft.lbs.)	Weight per 100 (lbs.)
Size in Inches	C	D	Thd.	G	N				
3/16 - 1/4	0.40	0.94	3/8-16	1.41	1.44	2	4.00	30	21
5/16	0.47	1.06	3/8-16	1.50	1.54	2	5.00	30	28
3/8	0.51	1.06	7/16-14	1.84	1.78	2	5.25	45	38
7/16 - 1/2	0.59	1.25	1/2-13	2.21	2.15	3	11.00	65	63
9/16 - 5/8	0.72	1.50	5/8-11	2.72	2.57	3	13.50	130	115
3/4	0.86	1.81	3/4-10	2.94	2.67	3	16.00	225	180
7/8	0.97	2.12	3/4-10	3.31	2.86	4	26.00	225	230
1	1.13	2.25	3/4-10	3.72	3.06	5	37.00	225	275

* Based on clean, unlubricated threads. The table above shows the minimum torque required to reach maximum holding power of wire rope clips.

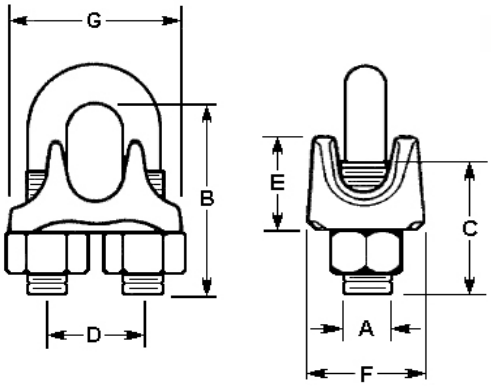
Meets the performance requirements of Federal Specification FF-C-450, current revision, Type 3, Class 1.

Improper use and installation of forged wire rope clips can result in serious injury or death.

Forged wire rope clips are the only type of wire rope clip approved for lifting purposes.

For complete installation instructions refer to page 48.

MALLEABLE WIRE ROPE CLIPS



ZINC PLATED, MALLEABLE

Size in Inches	Min. Clips Required	Wire Rope Turn Back in Inches	Torque in Foot Pounds*	Approx. Wt. in Pounds	Dimensions in Inches						
					A	B	C	D	E	F	G
1/16**	3	4	2.0	0.030	0.15	0.65	0.45	0.38	0.38	0.45	0.69
1/8**	3	4.3/4	3.0	0.040	0.18	0.81	0.50	0.50	0.50	0.56	0.94
3/16	3	5.1/2	4.5	0.063	0.25	0.94	0.56	0.56	0.56	0.63	1.06
1/4	3	7	15.0	0.130	0.31	1.19	0.75	0.75	0.69	0.75	1.31
5/16	3	7.3/4	15.0	0.150	0.31	1.31	0.84	0.75	0.75	0.75	1.44
3/8	3	9.1/2	30.0	0.210	0.38	1.63	1.00	0.88	0.84	0.88	1.63
7/16	4	10.1/4	40.0	0.370	0.38	2.00	1.19	1.06	1.00	1.06	1.88
1/2	4	15.1/4	45.0	0.370	0.44	2.00	1.19	1.06	1.00	1.06	1.88
9/16	4	16	50.0	0.590	0.50	2.31	1.38	1.25	1.25	1.28	2.09
5/8	4	16	75.0	0.590	0.50	2.31	1.38	1.25	1.25	1.28	2.09
3/4	5	22.1/4	75.0	0.840	0.56	2.56	1.56	1.31	1.44	1.56	2.38
7/8	5	23.1/2	130.0	1.250	0.63	3.06	1.81	1.63	1.75	1.81	2.88
1	6	31	130.0	1.660	0.63	3.44	2.00	1.88	2.06	2.00	3.00
1.1/8	7	39	200.0	2.430	0.75	4.00	2.75	2.00	2.19	2.06	3.38

Meets the performance requirements of Federal Specification FF-C-450, current revision, Type 1, Class 2.**

* Based on clean, unlubricated threads. The table above shows the minimum torque required to reach maximum holding power of wire rope clips.

** NOTE: 1/16" and 1/8" are not covered by Federal Specification FF-C-450, current revision.

Improper use and installation of cast malleable wire rope clips can result in serious injury or death.

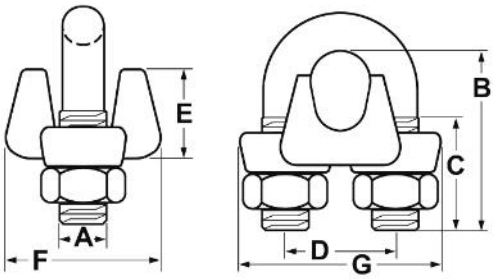
NEVER use cast malleable wire rope clips for lifting or suspending ANY load.

Cast malleable wire rope clips are to be used <u>only for non-critical, light duty applications</u> with small applied loads.

For complete installation instructions refer to page 48.

WARNING: NEVER EXCEED WORKING LOAD LIMIT
Read important warnings and information on pages 2 - 9 and page 48.

DROP FORGED WIRE ROPE CLIPS



GALVANIZED, DROP FORGED

Size in Inches	Min. Clips Required	Wire Rope Turn Back in Inches	Torque in Foot Pounds*	Approx. Wt. in Pounds	Dimensions in Inches						
					A	B	C	D	E	F	G
1/8**	2	3.1/4	4.5	0.05	0.19	0.75	0.44	0.44	0.38	0.81	0.94
3/16**	2	3.3/4	7.5	0.10	0.25	0.94	0.56	0.56	0.50	0.94	1.19
1/4	2	4.3/4	15.0	0.18	0.31	1.06	0.56	0.75	0.63	1.19	1.50
5/16	2	5.1/4	30.0	0.31	0.38	1.44	0.75	0.88	0.75	1.31	1.69
3/8	2	6.1/2	45.0	0.46	0.44	1.50	0.75	1.00	0.88	1.56	1.94
7/16	3	7	65.0	0.73	0.50	1.88	1.00	1.19	1.00	1.81	2.31
1/2	3	11.1/2	65.0	0.73	0.50	1.88	1.00	1.19	1.06	1.81	2.31
9/16	3	12	95.0	0.96	0.56	2.38	1.25	1.31	1.13	2.06	2.50
5/8	3	12	95.0	1.00	0.56	2.38	1.25	1.31	1.25	2.06	2.50
3/4	4	18	130.0	1.50	0.63	2.75	1.38	1.50	1.38	2.31	2.75
7/8	4	19	225.0	2.44	0.75	3.19	1.44	1.81	1.69	2.63	3.31
1	5	26	225.0	2.70	0.75	3.63	1.75	1.88	1.75	2.63	3.47
1.1/8	6	34	225.0	3.10	0.75	4.00	2.00	2.00	1.88	2.81	3.56
1.1/4	7	44	360.0	4.60	0.88	4.38	2.25	2.31	2.06	3.25	4.06
1.3/8	7	44	360.0	5.20	0.88	4.63	2.31	2.38	2.25	3.44	4.25
1.1/2	8	54	360.0	5.90	0.88	4.94	2.38	2.63	2.50	3.50	4.38
1.5/8	8	58	430.0	7.34	1.00	5.31	2.62	2.75	2.66	3.61	4.75
1.3/4	8	61	590.0	9.80	1.13	5.88	2.75	3.06	2.88	3.75	5.25
2	8	71	750.0	13.75	1.25	6.50	3.00	3.31	3.38	4.44	5.81
2.1/4	8	73	750.0	15.70	1.25	7.13	3.31	3.88	3.81	4.50	6.31
2.1/2	9	84	750.0	17.90	1.25	7.75	3.38	4.13	4.25	4.50	6.44
2.3/4	10	100	750.0	22.00	1.25	8.31	3.50	4.36	5.00	5.00	6.75
3	10	106	1,200.0	32.00	1.50	9.19	3.88	4.75	5.25	5.31	7.56

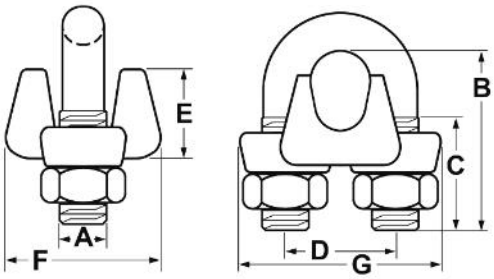
Meets the performance requirements of Federal Specification FF-C-450, current revision, Type 1 Class 1.
* Based on clean, unlubricated threads. The table above shows the minimum torque required to reach maximum holding power of wire rope clips.
** U-Bolts and nuts zinc plated.

Improper use and installation of forged wire rope clips can result in serious injury or death.

Forged wire rope clips are the only type of wire rope clip approved for lifting purposes.

For complete installation instructions refer to page 48.

DROP FORGED WIRE ROPE CLIPS



GALVANIZED, DROP FORGED

Size in Inches	Min. Clips Required	Wire Rope Turn Back in Inches	Torque in Foot Pounds*	Approx. Wt. in Pounds	Dimensions in Inches						
					A	B	C	D	E	F	G
1/8**	2	3.1/4	4.5	0.05	0.19	0.75	0.44	0.44	0.38	0.81	0.94
3/16**	2	3.3/4	7.5	0.10	0.25	0.94	0.56	0.56	0.50	0.94	1.19
1/4	2	4.3/4	15.0	0.18	0.31	1.06	0.56	0.75	0.63	1.19	1.50
5/16	2	5.1/4	30.0	0.31	0.38	1.44	0.75	0.88	0.75	1.31	1.69
3/8	2	6.1/2	45.0	0.46	0.44	1.50	0.75	1.00	0.88	1.56	1.94
7/16	3	7	65.0	0.73	0.50	1.88	1.00	1.19	1.00	1.81	2.31
1/2	3	11.1/2	65.0	0.73	0.50	1.88	1.00	1.19	1.06	1.81	2.31
9/16	3	12	95.0	0.96	0.56	2.38	1.25	1.31	1.13	2.06	2.50
5/8	3	12	95.0	1.00	0.56	2.38	1.25	1.31	1.25	2.06	2.50
3/4	4	18	130.0	1.50	0.63	2.75	1.38	1.50	1.38	2.31	2.75
7/8	4	19	225.0	2.44	0.75	3.19	1.44	1.81	1.69	2.63	3.31
1	5	26	225.0	2.70	0.75	3.63	1.75	1.88	1.75	2.63	3.47
1.1/8	6	34	225.0	3.10	0.75	4.00	2.00	2.00	1.88	2.81	3.56
1.1/4	7	44	360.0	4.60	0.88	4.38	2.25	2.31	2.06	3.25	4.06
1.3/8	7	44	360.0	5.20	0.88	4.63	2.31	2.38	2.25	3.44	4.25
1.1/2	8	54	360.0	5.90	0.88	4.94	2.38	2.63	2.50	3.50	4.38
1.5/8	8	58	430.0	7.34	1.00	5.31	2.62	2.75	2.66	3.61	4.75
1.3/4	8	61	590.0	9.80	1.13	5.88	2.75	3.06	2.88	3.75	5.25
2	8	71	750.0	13.75	1.25	6.50	3.00	3.31	3.38	4.44	5.81
2.1/4	8	73	750.0	15.70	1.25	7.13	3.31	3.88	3.81	4.50	6.31
2.1/2	9	84	750.0	17.90	1.25	7.75	3.38	4.13	4.25	4.50	6.44
2.3/4	10	100	750.0	22.00	1.25	8.31	3.50	4.36	5.00	5.00	6.75
3	10	106	1,200.0	32.00	1.50	9.19	3.88	4.75	5.25	5.31	7.56

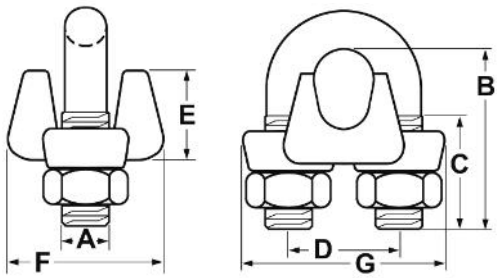
Meets the performance requirements of Federal Specification FF-C-450, current revision, Type 1 Class 1.
* Based on clean, unlubricated threads. The table above shows the minimum torque required to reach maximum holding power of wire rope clips.
** U-Bolts and nuts zinc plated.

Improper use and installation of forged wire rope clips can result in serious injury or death.

Forged wire rope clips are the only type of wire rope clip approved for lifting purposes.

For complete installation instructions refer to page 48.

STAINLESS STEEL WIRE ROPE CLIPS



STAINLESS STEEL, DROP FORGED

Size in Inches	Minimum Clips Required	Torque in Foot Pounds*	Approximate Weight in Pounds
1/8	2	4.5	0.050
3/16	2	7.5	0.106
1/4	2	15.0	0.190
5/16	2	30.0	0.300
3/8	2	45.0	0.460
1/2	3	65.0	0.680
5/8	3	95.0	1.000
3/4	4	130.0	1.500
7/8	4	225.0	2.440
1	5	225.0	2.700

Dimensions for drop forged stainless steel clips are the same as for the drop forged galvanized clips on page 51.
* Based on clean, unlubricated threads. The table above shows the minimum torque required to reach maximum holding power of wire rope clips.

Improper use and installation of forged wire rope clips can result in serious injury or death.

Forged wire rope clips are the only type of wire rope clip approved for lifting purposes.

For complete installation instructions refer to page 48.